2、实现 image 的霍夫曼编码

```
Image = [8 3 6 2 5 8 6 1;6 8 8 8 8 7 8 8;3 8 8 6 8 8 6 5; 6 3 5 8 4 3 8 6;
2 8 7 6 8 7 8 3;4 4 8 4 6 8 5 2;8 3 2 8 4 4 8 6; 1 4 8 6 8 3 8 5];
[h,w] = size(Image);
totalpixelnum = h*w;
len = max(Image(:)) + 1;
for graynum = 1:len
    gray(graynum,1) = graynum - 1;
end
for graynum = 1:len
    histgram(graynum) = 0;
    gray(graynum, 2) = 0;
    for i = 1:w
        for j = 1:h
            if gray(graynum,1) == Image(j,i)
                histgram(graynum) = histgram(graynum) + 1;
            end
        end
    end
    histgram(graynum) = histgram(graynum) / totalpixelnum;
end
histbackup = histgram;
sum = 0;
treeindex = 1;
while(1)
    if sum>= 1.0
        break;
    else
        [sum1, pl] = min(histgram(1:len)) ;
        histgram (pl) = 1.1;
        [sum2, p2] = min(histgram(1:len)) ;
        histgram(p2) = 1.1;
        sum = sum1 + sum2 ;
        len = len + 1;
        histgram(len) = sum;
        tree(treeindex, 1) = pl;
        tree(treeindex, 2) = p2 ;
        treeindex = treeindex + 1;
    end
end
for k = 1:treeindex - 1
    i = k;
    codevalue = 1;
    if or(tree(k, 1) <= graynum, tree(k, 2) <= graynum)
```

```
            if tree(k, 1) <= graynum
                gray(tree(k,1),2) = gray(tree(k,1),2) + codevalue;
                codelength = 1;
                while( i < treeindex - 1)
                    codevalue = codevalue * 2;
                    for j = i:treeindex - 1
                        if tree(j , 1) == i + graynum
                            gray(tree(k,1),2) = gray(tree(k,1),2) + codevalue;
                            codelength = codelength + 1;
                            i = j;
                            break;
                        elseif tree(j, 2) == i + graynum
                            codelength = codelength + 1;
                            i = j;
                            break;
                        end
                    end
                end
                gray(tree(k,1) ,3) = codelength;
            end
            i = k;
            codevalue = 1;
            if tree(k, 2) <= graynum
                codelength = 1;
                while(i < treeindex - 1)
                    codevalue = codevalue*2;
                    for j = i:treeindex - 1
                        if tree(j , 1) == i + graynum
                            gray(tree(k,2),2) = gray(tree(k,2),2) + codevalue;     codelength
= codelength + 1;
                            i = j;
                            break;
                        elseif tree(j,2) == i + graynum
                            codelength = codelength + 1;
                            i = j;
                            break;
                        end
                    end
                end
                gray(tree(k,2),3) = codelength;
            end
        end
    end
    for k = 1:graynum
```

```
        RstImg{k} = dec2bin(gray(k,2),gray(k, 3)) ;
end
disp('霍夫曼编码');
disp(RstImg);
```